6.1.1 Memberships in Profession Societies at National/International Levels (5)

Table No. 6.1.1.1: List of faculty members and their memberships.

S.No	Name of the Faculty	Name of the Professional Society /Body at National and International Level	Name of the Grade/ Level/Posit
1	Dr. Saurav Chakraborty	Indian Society of Mechanical Engineers	Life Member
2	Dr. Saurav Chakraborty	American Society of Mechanical Engineers	Member
3	Dr. Sushil Kumar Dhiman	Society of Automobile Engineers India	Member
4	Dr. Dipti Prasad Mishra	Institution of Engineers India	Life Member
5	Dr. Dipti Prasad Mishra	Indian Institution of Industrial Engineering	Life Member
6	Dr. Dipti Prasad Mishra	Indian Society for Technical Education	Life Member
7	Dr. Sujeet Kumar Mishra	Institution of Engineers India	Fellow
8	Dr. Anil Chandra Mahato	Fluid Power Society of India	Life Member
9	Dr. Sujeet Kumar Mishra	The Indian Science Congress Association	Life Member
10	Dr. Sujeet Kumar Mishra	Materials Research Society of India	Life Member
11	Dr. Sujeet Kumar Mishra	Association for Machines and Mechanism	Life Member
12	Dr. Apurba Kumar Roy	Fluid Power Society of India	Life Member
13	Dr. Apurba Kumar Roy	Institution of Engineers India	Life Member
14	Dr. Apurba Kumar Roy	Indian Society for Technical Education	Life Member
15	Dr. Praveen Mishra	Indian Society for Technical Education	Life Member
16	Dr. Praveen Mishra	Institution of Engineers India	Fellow
17	Dr. Om Prakash	Indian National Academy of Engineering	Senior Member
18	Dr. O. P. Pandey	Institution of Engineers India	Life Member
19	Dr. Rajeev Kumar	Indian Society of Technical Education	Life Member
20	Dr. Rajeev Kumar	Institution of Engineers India	Life Member
21	Dr Richa Pandey	Indian Society for Technical Education	Life Member
22	Dr Richa Pandey	Geriatric Society of India	Life Member
23	Dr Richa Pandey	International Society of Prosthetics and Orthodics	Life Member
24	Dr. Paritosh Mahata	Society of Automotive Engineers India	Member
25	Dr. Lakhbir Singh Brar	Indian Society of Technical Education	Life Member
26	Dr Richa Pandey	Artificial Intelligence and Machine Learning Association	Life Member
27	Dr. Mukesh Sharma	Indian Society of Heating, Refrigerating and Air Conditioning Engineers	Member

6.1.2 Faculty as Resource Persons or Participants in STTPs/FDPs (10)

6.1.2.1 Faculty as Resource Persons in STTPs/FDPs (5)

Table No. 6.1.2.1: List of faculty members as resource person in STTP/FDP events.

(CAYm1) 2024-25

S.No	Name of the Faculty as Resource Person	Name of the STTP/FDP	Date (DD/MM/YYYY)	Location	Organized by
1	Dr. Sushanta Ghuku	Innovative Synergies in Mechanical Engineering-Design to Sustainability	21/04/2025	Sanaka Educational Trust's Group of Institution, Durgapur	Sanaka Educatior Durgapur
2	Dr.Sushil Kumar Dhiman	Emerging Trends in Advanced Materials Manufacturing and Surface Engineering	09/12/2025	NIT Jamshedpur	NIT Jamshedpur
3	Dr. Om Prakash	ANRF sponsored Workshop on Green Hydrogen	20/01/2025	BIT Mesra, Ranchi	BIT Mesra, Ranch
4	Dr. Anil Chandra Mahato	Intelligent Monitoring and Optimization of Machining/Finishing Processes using Industry 4.0 Technolo	07/08/2025	Thapar Institute of Engineering and Technology, Patiala	Thapar Institute o Patiala
5	Dr. Dipti Prasad Mishra	Advanced Heat Transfer Technologies for Enhancing Energy Efficiency and Reducing Carbon Emission Glo	13/01/2025	NIT Jamshedpur	NIT Jamshedpur
6	Dr.Sushil Kumar Dhiman	Modern Tools & Techniques for Surface Modification	15/09/2025	BIT Mesra, Ranchi	BIT Mesra, Ranch
7	Prof. Kaushik Kumar	Special Lecture on 5S	23/07/2024	Ranchi, Jharkhand	Jharkhand Goverr
8	Prof. Kaushik Kumar	SVU Faculty Lecture Series	24/12/2024	Kolkata, West Bengal	Swami Vivekanan
9	Prof. Kaushik Kumar	AICTE Sponsored FDP on Recent Advances in Business Analytics and Management Research	08/01/2025	Ranchi, Jharkhand	IIIT Ranchi

(CAYm2) 2023-24

S.No	Name of the Faculty as Resource Person	Name of the STTP/FDP	Date (DD/MM/YYYY)	Location	Organized by
1	Prof. Kaushik Kumar	Professional Approach on Mould Design	15/03/2023	Ranchi, Jharkhand	Jharkhand Governmen
2	Prof. Kaushik Kumar	Seminar and Hands-on Training on Machining	20/04/2023	Ranchi, Jharkhand	Usha Martin University,
3	Prof. Kaushik Kumar	Workshop on Paper Writing – The State of Art	04/08/2023	Ranchi, Jharkhand	RTC Institute of Techno
4	Prof. Kaushik Kumar	Expert Lecture on Effective Research Paper Writing – Augmenting Students' Employability	10/08/2023	Ranchi, Jharkhand	RKDF University, Ranch
5	Prof. Kaushik Kumar	Workshop on Intellectual Property Right	11/08/2023	Ranchi, Jharkhand	RTC Institute of Techno
6	Prof. Kaushik Kumar	Expert Lecture on Additive Manufacturing	11/10/2023	Hyderabad, Telangana	GRIET, Hyderabad
7	Prof. Kaushik Kumar	Workshop on Intellectual Property Right	04/11/2023	Ramgarh, Jharkhand	Radha Govind Universi

(CAYm3) 2022-23

S.No	Name of the Faculty as Resource Person	Name of the STTP/FDP	Date (DD/MM/YYYY)	Location	Organized by
1	Dr. Ritwik Maiti	Preparation and Characterization of Oral Solid Dosage Form	10/07/2022	Birla Institute of Technology Mesra, Ranchi	ISM Dhanbad and
2	Prof. Kaushik Kumar	Skilling Workshop on 3D Printing	14/12/2022	AICTE IDEA Lab, BIT Mesra, Ranchi	BIT Mesra, Ranch
3	Prof. Kaushik Kumar	Workshop on Effective Research Paper Writing	03/09/2022	Ranchi, Jharkhand	Usha Martin Univ
4	Prof. Kaushik Kumar	Workshop on Project Proposal Writing	30/07/2022	Ranchi, Jharkhand	Cambridge Institu

6.1.2.2 Faculty Members' Participation in STTPs/FDPs (5)

Name of the faculty	Max 5 Per Faculty		
	2024-25(CAYm1)	2023-24(CAYm2)	2022-23(CAYm3)
Dr. Saurav Chakraborty	3.00	0.00	0.00
Dr. Sujeet Kumar Mishra	5.00	5.00	0.00
Dr. Apurba Kumar Roy	0.00	5.00	0.00
Dr. Om Prakash	5.00	0.00	0.00
Dr. Mukesh Sharma	5.00	0.00	0.00
Dr. L S Brar	3.00	0.00	0.00
Dr. Arkadeb Mukhopadhyay	3.00	3.00	0.00
Sum	24.00	10.00	0.00
RDF = Number of faculty required to comply with the 20:1 student - faculty ratio in the Department alone, as per section 5.1 of SAR(RDF= DS / 20).	20.40	20.40	20.40
Assessment Points (AP)= (Sum/(0.5* RDF)) (Points limited to 5 for each assessment year)	2.35	0.98	0.00

Average assessment over 3 years: 1.11

6.1.3 Faculty Contribution in Development of SWAYAM MOOCs and other E-Content (7)

Table No. 6.1.3.1: List of faculty members developed MOOC course for the past 3 years.

S.No	Name of the Faculty	Name of the Course Developed and available online on Swayam platform by your Department faculty
1		

6.1.4 Faculty Certification of MOOCs through SWAYAM, etc. (8)

Table No. 6.1.4.1: List of faculty members obtained certification of MOOCs for the past 3 years.

S.No	Name of the Faculty	Name of Course Passed	Course Offered by (agency)	Grade obtained if any
1				

6.1.5 FDP/STTP Organized by the Department (10)

Table No. 6.1.5.1: List of FDPs/STPs organized by Department for the past 3 years.

(CAYm1) 2024-25

S.No	Name of the Program	Date of the Program(DD/MM/YYYY)	Duration	Name of the Speaker & Designation and Organization
1	Green Hydrogen: The Future of Clean Energy	20/01/2025	5 days	Dr. Anil Kumar, Professor, Delhi Technical University, New Delhi, and many others

(CAYm2) 2023-24

S.No	Name of the Program	Date of the Program(DD/MM/YYYY)	Duration	Name of the Speaker & Designation and Organization
1	7 Day Karyashala on Surface Engineering and Tribological Challenges in Sustainable Manufacturing	18/12/2023	7 days	Dr. Partha Protim Chattopadhyay, Professor & Director, NIAMT Ranchi and many o

(CAYm3) 2022-23

S.No	Name of the Program	Date of the Program(DD/MM/YYYY)	Duration	Name of the Speaker & Designation and Organization
1	7 Day Karyashala on Joining techniques for modern transportation sector	5/12/2022	7 days	Prof. Surjya K Pal, Professor IIT Kharagpur and others

6.1.6 Faculty Support in Student Innovative Projects (10)

Table No. 6.1.6.1: List of faculty members involved in student innovative projects.

(CAYm1) 2024-25

S.No	Name of the Faculty	Name of the Event	Date of the Event(DD/MM/YYYY)	Place of Event	Website Link if a
1	Dr. Paritosh Mahata	BAJA SAE INDIA 25	09-01-2025	NATRAX, Pithampur, Madhya Pradesh	https://www.bajas

(CAYm2) 2023-24

S.No	Name of the Faculty	Name of the Event	Date of the Event(DD/MM/YYYY)	Place of Event	Website Link if a
1	Dr. Paritosh Mahata	BAJA SAE INDIA 24	10/01/2024	NATRAX, Pithampur, Madhya Pradesh	https://www.bajas

(CAYm3) 2022-23

S.No	Name of the Faculty	Name of the Event	Date of the Event(DD/MM/YYYY)	Place of Event	Website Link if a
1	Dr. Paritosh Mahata	BAJA SAE INDIA 23	15/02/2023	NATRAX, Pithampur, Madhya Pradesh	https://www.bajas

6.1.7 Faculty Internship/Training/Collaboration with Industry (10)

Table No. 6.1.7.1: Faculty internship/training/collaboration details.

S.No	Name of the Faculty	Name of the Internship/ Training/ Collaboration	Name of the Company & Place	Duration	Outcomes of Internship/ Training/ Collaborati
1	Dr. Saurav Chakraborty	Research Collaboration	Tata Steel Pvt. Ltd., Jamshedpur	2022-2023	1 SCI journal paper (https://doi.org/10.1080/014
2	Dr. Om Prakash, Dr. Nirmal Kumar, Dr. Lakhbir Singh Brar	Research Collaboration	MIR Lab, Moradabad, U.P.	2022-2025	Product development
3	Dr. Rajeev Kumar	Research Collaboration	Aman Infra, Ranchi	2022-2025	Stirling Engine Prototype development

6.2 Research and Development Activities (60)

6.2.1 Academic Research (10)

Table No. 6.2.1.1: Faculty publication details.

S.No.	Item	2024-25 (CAYm1)	2023-24 (CAYm2)	2022-23 (CAYm3)
1	No. of peer reviewed journal papers published	65	73	52
2	No. of peer reviewed conference papers published	25	24	26
3	No. of books/book chapters published	8	6	5

6.2.2 Ph.D. Student Details (5)

Table No. 6.2.2.1: Ph.D. details.

S.No.	Item	2024-25 (CAYm1)	2023-24 (CAYm2)	2022-23 (CAYm3)
1	No. of students enrolled for Ph.D. in the Department	5	4	3
2	No. of Ph.D. students graduated in the Department	5	7	0

6.2.3 Development Activities (10)

S.No	Name of Faculty	Details of Patents	CAYm1 (24-25)		CAYm2 (23-24)		CAYm3 (22-23)	
		Title	No.s	Status (Published/Granted)	No.s	Status (Published/Granted)	No.s	Status (Published/Granted)
1	Dr. Lakhbir Singh Brar	Hydrogen tank for usage in 2/3 wheelers (HydroRide)		Published				
2	Dr. Lakhbir Singh Brar	A fin and tube heat exchanger with inclined continuous vortex				Published		
3	Dr. Lakhbir Singh Brar	A fin and tube heat exchanger				Published		
4	Dr. Lakhbir Singh Brar	A heat storage and release system for energy management in solar power plant				Published		
5	Dr. Lakhbir Singh Brar	A fin and tube heat exchanger				Published		

6	Dr. Mukesh Sharma	A MACHINE LEARNING BASED INTELLIGENT DUST SUPPRESSION SYSTEM		Published
7	Dr. Mukesh Sharma	DUST COLLECTION AND SUPPRESSION DEVICE		Published
8	Dr. Mukesh Sharma	DEVICE TO DETERMINE THICKNESS OF WATER FILM DURING OIL - WATER FLOW	Published	
9	Dr. Mukesh Sharma	A SOLAR STILL APPARATUS AND METHOD OF USE THEREOF	Published	

10	Dr. D.P. Mishra	A Fin and Tube Heat Exchanger		Published
11	Dr. D.P. Mishra	A Fin and Tube Heat Exchanger with Inclined Continuos Vertex Generato		Published
12	Dr.Om Prakash	Hydrogen tank for usage in 2/3 wheelers (HydroRide)	Published	
13	Dr.Om Prakash	A fin and tube heat exchanger with inclined continuous vortex		Published
14	Dr.Om Prakash	A fin and tube heat exchanger		Published

15	Dr.Om Prakash	A heat storage and release system for energy management in solar power plant		Published
16	Dr.Om Prakash	A fin and tube heat exchanger		Published
17	Dr. Anil C. Mahato	A variable inertia flywheel for power controlling and delivering systems		Published on January 17, 2024.
18	Dr. Anil C. Mahato	A wind turbine power transmission system	Published on April 04, 2025	
19	Dr. Anil C. Mahato	Variable inertia flywheel system for inverted pendulum control	Publishedon June 18, 2025.	
20	Dr. Rajeev Kumar	An Alpha Steam –based Stirling System for Power Generation and Method thereof	Published on 05.04.2024	

21	Dr. Rajeev Kumar	A Modified Plant- Based Polysaccharide Binder, Methods of its Preparation and Use thereof		Published on 15.12.2023
22	Dr. Arkadeb Mukhopadhyay	A metal coating process for tribological applications and a coated surface obtained therefrom	1	Published
23	Dr. Arkadeb Mukhopadhyay	A surface coating and method of preparation thereof using computational tools	1	Published

Name of the Faculty	working models Title	CAYm1 (24-25)	CAYm2 (23-24)	CAYm3 (22-23)
Dr. Om Prakash	A heat storage and release system for energy management in solar power plant		Working	
Dr. Anil Mahato	variable inertia flywheel development			

Dr. Anil Mahato	Wind tubine power transmission system			
Dr. Anil Mahato	variable inertia flywheel in inverted pendulum			

Prototypes developed by faculty members

Name of Faculty	Title	CAYm1 (24-25)	CAYm2 (23-24)	CAYm3 (22-23)
Dr. Om Prakash	A heat storage and release system for energy management in solar power plant		Prototype	

6.2.4 Sponsored Research Project (15)

2024-25 (CAYm1)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration o project
Lakhbir Singh Brar	Om Prakash, Arkadeb Mukhopadhayay	Mechanical Engineering	Development of abrasion-resistant electroless Ni-B-W coating on cyclone separator walls and design optimization for longevity with enhanced performance	ANRF	2024-2027
Priyank Kumar	Sudip Das, Lakhbir Singh Brar	Space Engineering and Rocketry	Development of High Angle of Attack CFD Aerodynamic Database & Behaviour Model of Missile Configuration at Varying Geometric and Flow Parameters	DRDO	2024-2026
Om Prakash	NA	Mechanical Engineering	Green Hydrogen: The Future of Clean Energy	ANRF	2024-2025

2023-24 (CAYm2)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of project
Dr. Sushanta Ghuku	NA	Mechanical Engineering	A novel hard magnetic soft continuum curved robot: Multi-physical mechanics, active control, modulation, and optimized design (Acronym: NHMSC robot)	ANRF	2024-2027

2022-23 (CAYm3)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration project
Om Prakash	Lakhbir Singh Brar, D.P.Mishra, D.Singh, Anil Kumar	Mechanical Engineering	Investigation of Stones as Sensible Thermal Heat Storage Materials for Concentrated Solar Power Applications	ANRF	2022-202
Rajeev Kumar	Richa Pandey	Mechanical Engineering	Design and Fabrication of Bellow Type of Bio-gas Plant	MoE, Unnat Bharat Abhiyan	2022-202
Rajeev Kumar	Rajeshwari Chatterjee	Mechanical Engineering	Setting up of a small-scale pilot plant for production of organic compost from kitchen waste and dry leaves	Unnat Bharat Abhiyan	2022-202
Rajeev Kumar	Sanath Kumar Mukherjee	Mechanical Engineering	SETTING UP OF A SMALL-SCALE PILOT PLANT FOR PRODUCTION OF BIODIESEL FROM INDIGENOUS PLANT	Unnat Bharat Abhiyan	2022-202
Arkadeb Mukhopadhyay	Arun Kumar Kadian, Anil C Mahato, Mukesh Sharma	Mechanical Engineering	Development of multi-layered lead free electroless Ni-B coating with high wear resistance, thermal stability and corrosion resistance for tribological applications	ANRF	2023-202

Total Amount (Lacs) Received for the Past 3 Years: 180.40

Note*:

- Only sponsored research projects will be considered. Infrastructure-based projects will not be considered here.

6.2.5 Consultancy Work (15)

2024-25 (CAYm1)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of th project
Dr. Arbind Kumar	Dr. A.K.Roy	Mechanical Engineering	Design Verification of 1 HP DC Surface Movable Trolley	Royal Instrument India Pvt. Ltd, Ratu Road, Ranchi	6 months
Dr. Arbind Kumar	Dr. A.K.Roy	Mechanical Engineering	Design Verification of 1 HP and 2 HP Solar Mobile DC Pumping System	Royal Instrumen India Pvt. Ltd, Ratu Road, Ranchi	6 months
Dr. Arbind Kumar	Dr. A.K.Roy	Mechanical Engineering	Evaluation of the structural design of trolley for 2 HP Solar Movable pumps	Directorate of Agriculture, Jharkhand	6 months
Nirmal Kumar	Om Prakash	Mechanical Engineering	Material Testing	Executive Engineer, Drinking Water & Sanitation Department, Jhumari Tilaiya	5 days
Nirmal Kumar	Om Prakash	Mechanical Engineering	Material Testing	M/s NCC Limited, Water and Environmental Division, Ranchi	5 days

2023-24 (CAYm2)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project
Nirmal Kumar	Om Prakash	Mechanical Engineering	Material Testing	M/s Sharda Shelters Pvt. Ltd. Ranchi	5 days
Nirmal Kumar	Om Prakash	Mechanical Engineering	Material Testing	Executive Engineer, Drinking Water & Sanitation Department, Medininagar	5 days

2022-23 (CAYm3)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project
Nirmal Kumar	Om Prakash	Mechanical Engineering	Material Testing	M/s NCC Limited, Ranchi	5 days
Nirmal Kumar	Om Prakash	Mechanical Engineering	Material Testing	Basics Engineering India Structure Pvt. Ltd., Ranchi	5 days

Total amount (Lacs) received for the past 3 years: 2.14

Note*:

- Only consultancy projects will be considered. Infrastructure-based projects will not be considered here.

6.2.6 Institution Seed Money or Internal Research Grant to its Faculty for Research Work(5)

6.2.6 A Amount received (3)

2024-25 (CAYm1)

Faculty name	Project title/ Support for Activity	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25	Amount Utilized(Lacs) i.e. 15,25,000=15.25	Outcomes o

2023-24 (CAYm2)

Faculty name	Project title/ Support for Activity	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25	Amount Utilized(Lacs) i.e. 15,25,000=15.25	Outcomes of the project
Saurav Chakraborty	Experimental Investigation of Dynamics of Drop Impact on a Heated Surface	2 years	2.50	0.86	Experimental studies conducted and 1 c
			Amount received (Rs.): 2.50		

2022-23 (CAYm3)

Faculty name	Project title/ Support for Activity	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25	Amount Utilized(Lacs) i.e. 15,25,000=15.25
Nirmal Kumar	DESIGN AND DEVELOPMENT OF A HEART SIMULATOR TO TEST HEART VALVES	2 years	5.00	4.64
			Amount received (Rs.): 5.00	

Total amount (Lacs) received for the past 3 years : 7.50

6.2.6 B Amount utilized (2)

These budget is utilized to develop product and utilized amout is 5.5 Lakhs.

7 FACILITIES AND TECHNICAL SUPPORT (100)

7.1 Adequate and well equipped laboratories, and technical manpower (40)